

Fine-Timestep Three-Moment Calibration Audit

NLS numerical study

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Abstract

We re-evaluate the fine-timestep three-mode NESS estimator with exactly the three-moment exponential calibration required by the coarse replication. No neural weight is retrained and no trajectory, split, shrink factor, normalizer, or gate is changed. Fine validation and the declared final test pass, the equilibrium known-answer control passes, and the resulting three-moment fine/coarse centered log-density RMS remains 0.0244 against the frozen 0.10 threshold. The procedural asymmetry between timesteps is therefore removed.

1 Frozen procedure and test access

For each base seed 8201, 8202, 8203, the fixed neural correction is augmented by an exponential tilt in

$$I_1, \quad I_1 I_2 \sin \theta_1, \quad I_2 I_3 \sin \theta_3.$$

The unshrunk coefficient vector is fitted on exactly the already opened rows used by the prior fine calibration and then multiplied by 0.625. The new V6 validation and test trajectories do not enter the fit. Before any R11 value was read, the protocol authorized one additional and final test access only if driven and equilibrium validation both passed. Both passed; their hashes were then recorded before the test was opened. All prior two-moment artifacts are retained.

2 Blind-test diagnostics

Table 1: Per-seed driven test diagnostics. All thresholds are unchanged.

step/protocol	seed	ESS	FP med.	FP p90	max $ z $	max TV
fine, 2-moment	8201	0.916008	0.088747	0.406029	1.342781	0.024294
fine, 2-moment	8202	0.916508	0.095015	0.408698	1.443345	0.024401
fine, 2-moment	8203	0.917796	0.091415	0.396536	1.220313	0.024247
fine, 3-moment	8201	0.915647	0.088694	0.400716	1.368753	0.024319
fine, 3-moment	8202	0.916244	0.095482	0.413261	1.450155	0.024421
fine, 3-moment	8203	0.917330	0.091638	0.401961	1.232539	0.024276
coarse, 3-moment	8201	0.908708	0.091214	0.408893	2.829839	0.024090
coarse, 3-moment	8202	0.909603	0.089429	0.402181	2.694922	0.024052
coarse, 3-moment	8203	0.911359	0.093102	0.434454	2.651876	0.024189

Every fine three-moment gate passes. The maximum pairwise centered log-ratio RMS is 0.012645, versus 0.012733 for fine two-moment and 0.013834 for coarse three-moment.

3 Calibration coefficients and induced shift

Table 2: Shrunk standardized exponential-tilt coefficients.

base seed	two-moment vector	three-moment vector
8201	(−0.012283, 0.002684)	(−0.012219, 0.002679, −0.001261)
8202	(−0.007364, 0.003292)	(−0.007317, 0.003289, −0.000940)
8203	(−0.010441, 0.005026)	(−0.010357, 0.005023, −0.001690)

On 50000 deterministic fine test states, the ensemble centered log-density shift has RMS 0.001518, absolute-shift p90 0.002103, p99 0.006155, and maximum 0.023228. Including the separately estimated normalizers gives RMS 0.001518 and mean -2.16×10^{-6} . On coarse test support, the centered RMS is 0.001543. Thus the third calibration direction is measurable but small.

4 Timestep replication under an identical protocol

The three-moment fine/coarse ensemble centered log-density RMS is

$$0.024355 \quad \text{on fine support}, \quad 0.024353 \quad \text{on coarse support}.$$

Both pass the 0.10 gate. The previous two-versus-three-moment values were 0.024257 and 0.024272, so the change is 9.8×10^{-5} and 8.2×10^{-5} .

5 Equilibrium control and verdict

The same three-moment model class is used with zero base correction and zero calibration coefficients. On test, ESS is 1, FP median is 8.88×10^{-16} , FP p90 is 3.55×10^{-15} , maximum moment $|z| = 1.272210$, maximum TV is 0.033212, and seed disagreement is zero. This checks the operator, diagnostics, and sampled Gibbs control; because $f = 0$ is imposed, it does not test the learned nonequilibrium correction.

The R11 result is PASS. Both timesteps now use the identical three-moment, 0.625-shrunk calibration protocol. The warranted claim remains a numerical NESS density on sampled stationary support, not global control in unsampled tails or an analytic stationary solution.

Machine-readable evidence

Complete diagnostics, moment rows, marginal rows, seed comparisons, normalizers, coefficient tables, density-shift tables, source hashes, and exact commands are stored under `recovery_r11_fine/`, `R11_S0`, `URCE_HASHES.sha256`, and `R11_COMMANDS.txt`.